

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Wire armored and PVC Sheathed

Voltage Grade

12 / 20 (24) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum wire and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502 or BS 6622
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Aluminum Wires applied over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request

Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
													
mm²	Ω/km	Ω/km	Ω/km	µF/km	mH/km	mH/km	A	A	A	A	mm	kg/km	
A - Copper Conductors													
PM-CT14-X8-01-RP	50	0.3870	0.4937	0.4937	0.1836	0.5218	0.4756	228	220	244	241	32.7	1540
PM-CT15-X8-01-RP	70	0.2680	0.3420	0.3420	0.2087	0.4941	0.4479	275	268	303	301	34.7	1825
PM-CT16-X8-01-RP	95	0.1930	0.2465	0.2466	0.2294	0.4731	0.4269	323	318	366	365	36.2	2110
PM-CT17-X8-01-RP	120	0.1530	0.1957	0.1957	0.2486	0.4558	0.4096	360	358	419	418	37.8	2405
PM-CT18-X8-01-RP	150	0.1240	0.1588	0.1589	0.2692	0.4412	0.3950	394	398	473	468	39.3	2805
PM-CT19-X8-01-RP	185	0.0991	0.1273	0.1275	0.2937	0.4273	0.3811	443	445	539	530	41.3	3220
PM-CT20-X8-01-RP	240	0.0754	0.0973	0.0976	0.3429	0.4162	0.3700	501	503	628	623	45.0	4000
PM-CT21-X8-01-RP	300	0.0601	0.0782	0.0785	0.3547	0.4030	0.3568	545	556	712	689	47.2	4655
PM-CT22-X8-01-RP	400	0.0470	0.0620	0.0624	0.3912	0.3910	0.3448	592	614	808	768	50.3	5690
PM-CT23-X8-01-RP	500	0.0366	0.0494	0.0500	0.4357	0.3779	0.3317	646	675	914	851	53.8	6865
PM-CT24-X8-01-RP	630	0.0283	0.0396	0.0404	0.4909	0.3678	0.3216	700	735	975	995	58.1	8320
PM-CT25-X8-01-RP	800	0.0221	0.0326	0.0336	0.5500	0.3557	0.3095	743	791	1065	1106	62.7	10270
PM-CM26-X8-01-RP	1000	0.0176	0.0250	0.0252	0.6814	0.3535	0.3073	818	874	1216	1264	73.7	13500
B - Aluminum Conductors													
PM-AT14-X8-01-RP	50	0.6410	0.8219	0.8219	0.1892	0.5145	0.4683	180	171	191	188	33.1	1280
PM-AT15-X8-01-RP	70	0.4430	0.5682	0.5682	0.2101	0.4917	0.4454	218	209	236	234	34.8	1415
PM-AT16-X8-01-RP	95	0.3200	0.4106	0.4106	0.2322	0.4707	0.4245	258	249	285	284	36.4	1565
PM-AT17-X8-01-RP	120	0.2530	0.3247	0.3248	0.2500	0.4548	0.4085	289	282	331	326	37.9	1700
PM-AT18-X8-01-RP	150	0.2060	0.2645	0.2646	0.2733	0.4386	0.3924	320	313	374	370	39.6	1930
PM-AT19-X8-01-RP	185	0.1640	0.2109	0.2110	0.2978	0.4250	0.3788	357	353	428	424	41.6	2135
PM-AT20-X8-01-RP	240	0.1250	0.1610	0.1612	0.3249	0.4162	0.3700	402	404	503	500	45.0	2545
PM-AT21-X8-01-RP	300	0.1000	0.1292	0.1294	0.3547	0.4030	0.3568	442	450	574	561	47.2	2805
PM-AT22-X8-01-RP	400	0.0778	0.1010	0.1013	0.3912	0.3910	0.3448	489	505	661	639	50.3	3320
PM-AT23-X8-01-RP	500	0.0605	0.0793	0.0797	0.4357	0.3779	0.3317	539	565	760	722	53.8	3810
PM-AT24-X8-01-RP	630	0.0469	0.0624	0.0629	0.4922	0.3675	0.3212	605	625	833	840	58.2	4455
PM-AT25-X8-01-RP	800	0.0367	0.0501	0.0508	0.5500	0.3557	0.3095	663	691	942	956	62.7	5200
PM-AM26-X8-01-RP	1000	0.0291	0.3890	0.0391	0.6814	0.3535	0.3073	740	778	1087	1114	73.7	6890

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 12.7/22 (24) KV